04 – Final evaluation on the test set (Phase 7)

The test rows were opened **once**, in Phase 7, after every model choice had been fixed. This notebook does **not** open them again: it reads the results saved at that moment.

In [1]:
```
# Setup: work from the project root so the project code and configs are found
import os, sys, json, platform, warnings
from pathlib import Path
ROOT = Path.cwd().parent if Path.cwd().name == "notebooks" else Path.cwd()
os.chdir(ROOT); sys.path.insert(0, str(ROOT))
warnings.filterwarnings("ignore")
import numpy as np, pandas as pd
from IPython.display import Image, Markdown, display
pd.set_option("display.precision", 3); pd.set_option("display.width", 160);
pd.set_option("display.max_columns", 20)
from src.utils.config import load_config, resolve_path
REPORTS = resolve_path(load_config("project")["paths"]["reports"])
FIG = resolve_path(load_config("project")["paths"]["figures"])
print("Project folder:", ROOT)
print("Python", platform.python_version(), "| numpy", np.__version__, "|
pandas", pd.__version__)
```

```
Project folder: C:\AL_outcome_predictor
Python 3.12.0 | numpy 2.5.4 | pandas 3.0.6
```

1. Record of the single test-set opening

In [2]:
```
access = json.loads((REPORTS / "phase07" / "test_access.json").read_text("utf-
8"))
print("Opened at (UTC):", access["opened_at_utc"])
print("Test rows:", access["test_rows"])
print("Networks evaluated:", len(access["networks_evaluated"]))
```

```
Opened at (UTC): 2026-10-15T04:11:14+00:00
Test rows: 153
Networks evaluated: 85
```

2. Results for each target (value with 95 % bootstrap interval), all systems

In [3]:
```
for target in ("zscore", "zscore_band", "university_group", "degree_category"):
    display(Markdown(f"### {target}"))
    display(pd.read_csv(REPORTS / "phase07" / f"results_{target}.csv"))
```

zscore

	system	mae	rmse	r2	spearman
0	Autoencoder + MLP	0.199 [0.174, 0.225]	0.257 [0.226, 0.289]	0.648 [0.550, 0.725]	0.813 [0.741, 0.864]
1	Entity-embedding MLP	0.207 [0.180, 0.234]	0.268 [0.237, 0.301]	0.617 [0.513, 0.700]	0.803 [0.729, 0.857]
2	MLP	0.206 [0.180, 0.233]	0.266 [0.235, 0.297]	0.623 [0.523, 0.704]	0.802 [0.730, 0.856]
3	Multi-task	0.208 [0.184, 0.234]	0.266 [0.236, 0.295]	0.625 [0.522, 0.704]	0.805 [0.735, 0.857]
4	Regularised DNN	0.201 [0.177, 0.227]	0.259 [0.230, 0.290]	0.642 [0.552, 0.713]	0.817 [0.744, 0.867]
5	ResNet-style	0.210 [0.183, 0.237]	0.274 [0.240, 0.309]	0.601 [0.477, 0.692]	0.794 [0.716, 0.849]
6	TabNet	0.241 [0.211, 0.272]	0.309 [0.267, 0.350]	0.493 [0.340, 0.614]	0.701 [0.600, 0.784]
7	Wide & Deep	0.209 [0.184, 0.235]	0.266 [0.235, 0.297]	0.623 [0.524, 0.703]	0.799 [0.727, 0.850]
8	single_mlp_zscore	0.205 [0.180, 0.232]	0.263 [0.234, 0.293]	0.633 [0.534, 0.712]	0.800 [0.723, 0.855]
9	single_mtl_zscore	0.203 [0.178, 0.229]	0.259 [0.230, 0.288]	0.644 [0.550, 0.718]	0.807 [0.734, 0.860]
10	Stage-2 MLP (Z-score only)	0.205 [0.180, 0.232]	0.263 [0.234, 0.293]	0.633 [0.534, 0.712]	0.800 [0.723, 0.855]
11	Chosen system	0.205 [0.180, 0.232]	0.263 [0.234, 0.293]	0.633 [0.534, 0.712]	0.800 [0.723, 0.855]
12	Trivial baseline	0.367 [0.331, 0.404]	0.434 [0.400, 0.469]	-0.001 [-0.037, -0.000]	0.000 [0.000, 0.000]

zscore_band

	system	macro_f1	balanced_accuracy	accuracy	log_loss	brier	ece	roc_auc_ovr_ma
0	Autoencoder + MLP	0.708 [0.632, 0.772]	0.711 [0.643, 0.776]	0.712 [0.641, 0.778]	0.688 [0.580, 0.811]	0.412 [0.346, 0.479]	0.097 [0.056, 0.170]	0.856 [0.812, 0.8
1	Entity-embedding MLP	0.680 [0.603, 0.745]	0.679 [0.606, 0.746]	0.680 [0.601, 0.745]	0.728 [0.610, 0.856]	0.428 [0.358, 0.500]	0.066 [0.051, 0.149]	0.844 [0.796, 0.8
2	MLP	0.712 [0.641, 0.775]	0.711 [0.642, 0.777]	0.712 [0.647, 0.778]	0.737 [0.623, 0.851]	0.435 [0.367, 0.505]	0.079 [0.054, 0.162]	0.835 [0.785, 0.8
3	Multi-task	0.686 [0.612, 0.753]	0.685 [0.613, 0.753]	0.686 [0.608, 0.758]	0.716 [0.598, 0.841]	0.432 [0.357, 0.512]	0.077 [0.049, 0.154]	0.843 [0.793, 0.8
4	Regularised DNN	0.701 [0.628, 0.764]	0.704 [0.635, 0.769]	0.706 [0.634, 0.771]	0.680 [0.586, 0.785]	0.404 [0.342, 0.470]	0.048 [0.037, 0.136]	0.862 [0.815, 0.9
5	ResNet-style	0.700 [0.628, 0.769]	0.704 [0.637, 0.772]	0.706 [0.634, 0.778]	0.743 [0.633, 0.862]	0.434 [0.366, 0.506]	0.081 [0.052, 0.159]	0.836 [0.787, 0.8
6	TabNet	0.594 [0.522, 0.666]	0.594 [0.522, 0.666]	0.595 [0.523, 0.667]	0.875 [0.736, 1.019]	0.509 [0.437, 0.584]	0.087 [0.055, 0.166]	0.784 [0.729, 0.8
7	Wide & Deep	0.703 [0.630, 0.768]	0.704 [0.634, 0.770]	0.706 [0.634, 0.778]	0.719 [0.611, 0.831]	0.429 [0.360, 0.498]	0.065 [0.047, 0.147]	0.841 [0.793, 0.8
8	single_mlp_zscore_band	0.679 [0.608, 0.750]	0.684 [0.617, 0.755]	0.686 [0.621, 0.758]	0.730 [0.639, 0.822]	0.430 [0.371, 0.490]	0.118 [0.073, 0.186]	0.842 [0.794, 0.8
9	single_mtl_zscore_band	0.699 [0.629, 0.764]	0.698 [0.630, 0.767]	0.699 [0.634, 0.771]	0.717 [0.616, 0.821]	0.423 [0.357, 0.490]	0.043 [0.037, 0.132]	0.843 [0.794, 0.8
10	Chosen system	0.712 [0.641, 0.775]	0.711 [0.642, 0.777]	0.712 [0.647, 0.778]	0.737 [0.623, 0.851]	0.435 [0.367, 0.505]	0.079 [0.054, 0.162]	0.835 [0.785, 0.8
11	Trivial baseline	0.169 [0.141, 0.194]	0.333 [0.333, 0.333]	0.340 [0.268, 0.412]	1.099 [1.098, 1.099]	0.667 [0.666, 0.667]	0.006 [0.001, 0.084]	0.500 [0.500, 0.5

university_group

	system	macro_f1	balanced_accuracy	accuracy	log_loss	brier	ece	roc_auc_ovr
0	Autoencoder + MLP	0.270 [0.197, 0.338]	0.318 [0.226, 0.405]	0.275 [0.203, 0.346]	1.352 [1.272, 1.433]	0.720 [0.684, 0.754]	0.153 [0.106, 0.222]	0.737 [0.695
1	Entity-embedding MLP	0.335 [0.261, 0.400]	0.369 [0.286, 0.451]	0.373 [0.301, 0.444]	1.338 [1.251, 1.428]	0.700 [0.661, 0.741]	0.047 [0.036, 0.138]	0.732 [0.684
2	MLP	0.286 [0.212, 0.354]	0.339 [0.246, 0.428]	0.307 [0.235, 0.379]	1.384 [1.307, 1.464]	0.719 [0.685, 0.754]	0.088 [0.045, 0.160]	0.727 [0.679
3	Multi-task	0.340 [0.271, 0.410]	0.383 [0.301, 0.471]	0.379 [0.307, 0.458]	1.334 [1.251, 1.420]	0.695 [0.657, 0.732]	0.046 [0.029, 0.130]	0.741 [0.696
4	Regularised DNN	0.302 [0.229, 0.367]	0.402 [0.321, 0.481]	0.320 [0.248, 0.392]	1.339 [1.266, 1.415]	0.701 [0.669, 0.733]	0.098 [0.068, 0.177]	0.749 [0.704
5	ResNet-style	0.337 [0.259, 0.411]	0.366 [0.281, 0.452]	0.359 [0.287, 0.431]	1.338 [1.252, 1.419]	0.700 [0.662, 0.736]	0.087 [0.051, 0.165]	0.736 [0.684
6	TabNet	0.243 [0.177, 0.306]	0.300 [0.223, 0.378]	0.275 [0.203, 0.346]	1.418 [1.338, 1.508]	0.739 [0.704, 0.777]	0.122 [0.075, 0.191]	0.695 [0.644
7	Wide & Deep	0.337 [0.259, 0.410]	0.359 [0.268, 0.451]	0.366 [0.294, 0.438]	1.365 [1.270, 1.467]	0.705 [0.662, 0.749]	0.078 [0.045, 0.160]	0.723 [0.669
8	single_mlp_university_group	0.279 [0.215, 0.347]	0.289 [0.215, 0.367]	0.359 [0.288, 0.431]	1.412 [1.328, 1.495]	0.723 [0.686, 0.760]	0.028 [0.028, 0.121]	0.693 [0.637
9	single_mtl_university_group	0.284 [0.216, 0.353]	0.301 [0.221, 0.386]	0.327 [0.255, 0.399]	1.362 [1.281, 1.442]	0.707 [0.672, 0.743]	0.071 [0.040, 0.149]	0.723 [0.671
10	Chosen system	0.286 [0.212, 0.354]	0.339 [0.246, 0.428]	0.307 [0.235, 0.379]	1.384 [1.307, 1.464]	0.719 [0.685, 0.754]	0.088 [0.045, 0.160]	0.727 [0.679
11	Trivial baseline	0.106 [0.089, 0.122]	0.200 [0.200, 0.200]	0.359 [0.288, 0.438]	1.466 [1.391, 1.543]	0.745 [0.717, 0.773]	0.027 [0.000, 0.105]	0.500 [0.500

degree_category

	system	macro_f1	balanced_accuracy	accuracy	log_loss	brier	ece	roc_auc_ovr
0	Autoencoder + MLP	0.682 [0.602, 0.751]	0.721 [0.655, 0.785]	0.686 [0.608, 0.758]	0.858 [0.700, 1.027]	0.446 [0.370, 0.523]	0.106 [0.071, 0.180]	0.918 [0.894
1	Entity-embedding MLP	0.643 [0.566, 0.710]	0.672 [0.597, 0.743]	0.647 [0.575, 0.725]	0.919 [0.764, 1.077]	0.478 [0.400, 0.557]	0.091 [0.058, 0.165]	0.907 [0.880
2	MLP	0.622 [0.537, 0.696]	0.652 [0.575, 0.729]	0.634 [0.556, 0.712]	1.194 [1.063, 1.337]	0.580 [0.524, 0.639]	0.192 [0.123, 0.270]	0.883 [0.854
3	Multi-task	0.698 [0.623, 0.762]	0.729 [0.660, 0.793]	0.693 [0.621, 0.765]	0.906 [0.759, 1.066]	0.471 [0.395, 0.548]	0.128 [0.089, 0.204]	0.912 [0.886
4	Regularised DNN	0.680 [0.599, 0.750]	0.715 [0.646, 0.781]	0.686 [0.614, 0.758]	0.911 [0.762, 1.068]	0.471 [0.397, 0.548]	0.108 [0.073, 0.192]	0.915 [0.890
5	ResNet-style	0.649 [0.567, 0.715]	0.684 [0.611, 0.750]	0.647 [0.569, 0.719]	0.979 [0.839, 1.127]	0.481 [0.411, 0.554]	0.119 [0.084, 0.191]	0.905 [0.879
6	TabNet	0.508 [0.424, 0.585]	0.518 [0.433, 0.605]	0.510 [0.425, 0.595]	1.318 [1.128, 1.520]	0.627 [0.549, 0.708]	0.091 [0.069, 0.181]	0.834 [0.793
7	Wide & Deep	0.652 [0.578, 0.716]	0.689 [0.624, 0.751]	0.641 [0.562, 0.712]	1.013 [0.869, 1.169]	0.497 [0.426, 0.570]	0.095 [0.074, 0.174]	0.900 [0.872
8	single_mlp_degree_category	0.619 [0.537, 0.689]	0.649 [0.575, 0.720]	0.621 [0.542, 0.699]	1.013 [0.875, 1.152]	0.510 [0.442, 0.580]	0.111 [0.077, 0.186]	0.902 [0.875
9	single_mtl_degree_category	0.666 [0.585, 0.732]	0.692 [0.618, 0.763]	0.667 [0.595, 0.739]	0.865 [0.706, 1.033]	0.456 [0.375, 0.539]	0.067 [0.053, 0.161]	0.914 [0.887
10	Chosen system	0.622 [0.537, 0.696]	0.652 [0.575, 0.729]	0.634 [0.556, 0.712]	1.194 [1.063, 1.337]	0.580 [0.524, 0.639]	0.192 [0.123, 0.270]	0.883 [0.854
11	Trivial baseline	0.053 [0.040, 0.066]	0.143 [0.143, 0.143]	0.229 [0.163, 0.301]	1.881 [1.827, 1.937]	0.839 [0.824, 0.855]	0.001 [0.001, 0.077]	0.500 [0.500

3. Hypotheses H1–H6 against the success criteria locked in Phase 2

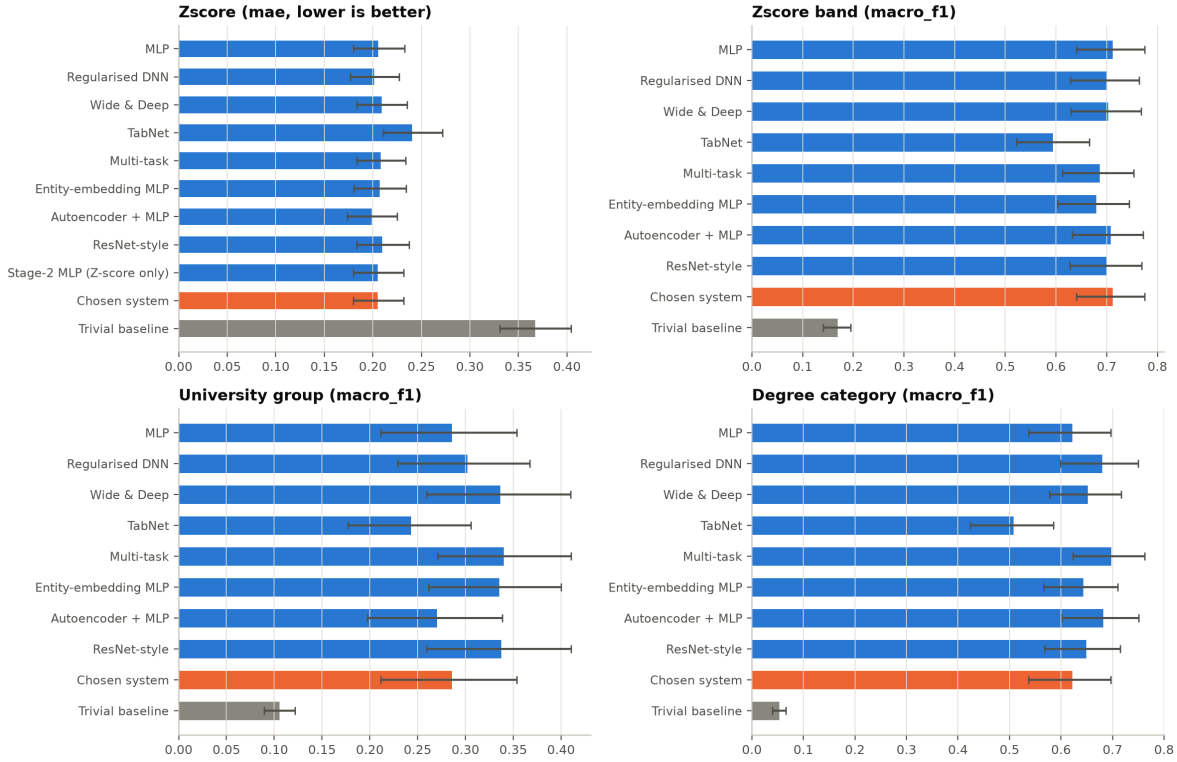
In [4]:
```
pd.read_csv(REPORTS / "phase07" / "hypotheses.csv")
```

Out[4]:

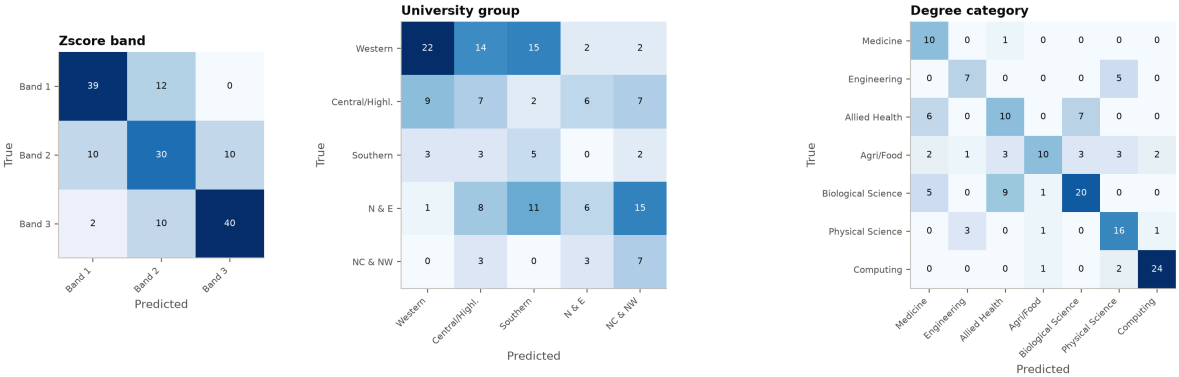
	hypothesis	target	result	evidence
0	H1	zscore	Supported	MAE reduction vs baseline: 0.442 [0.361, 0.515...
1	H2	zscore_band	Supported	macro F1 gain vs baseline: 0.543 [0.470, 0.615...
2	H3	university_group	Partly supported	macro F1 gain vs baseline: 0.180 [0.101, 0.249...
3	H4	degree_category	Supported	macro F1 gain vs baseline: 0.569 [0.480, 0.645...
4	H5	calibration	Not supported	zscore_band: ECE 0.079 [0.054, 0.162], BSS 0.3...
5	H6	multi-task vs single-task	Not supported (no difference detected)	mlp/zscore: no difference detected; mlp/zscore...

In [5]:
```
for name in ("comparison.png", "confusion_matrices.png", "reliability.png"):
    display(Image(str(REPORTS / "phase07" / name), width=760))
```

Results on the test set (seed 0, 95 % bootstrap CIs)



Chosen system: confusion matrices on the test set



Chosen system: reliability on the test set

